The Omega Axiom: Non-Lagrangian Wave-Based Derivation of Celestial Dynamics and Patterned Acceleration

Junho Ryu

2026-02-10

Abstract

This study reinterprets celestial acceleration, planetary magnetic fields, and atmospheric features as emergent phenomena from *compression* and *phase interference*. We provide a first-principles derivation of $F = ma$ based on the adsorption density of upper π-type waves and lower logarithmic stacking waves. Furthermore, we demonstrate that the 3-4-5 right triangle represents the fundamental threshold for material anchoring in a logarithmic phase-lattice. This framework enables deterministic, zero-latency control routines for autonomous systems by replacing probabilistic models with patterned phase-alignment.

1 Derivation of $F = ma$ via Wave Adsorption

1.1 The Duality of Wave and Material Anchoring

Under the Omega Axiom, material existence is not a collection of solid particles but an **Ordered Pattern** emerging from the interaction of two counter-propagating waves:

- **Upper Wave ($\Psi_{\mathbf{upper}}$)**: A π-type wave incoming from the external vacuum/universe, providing convergent compression.
- **Lower Wave ($\Psi_{\mathbf{lower}}$)**: A logarithmic stacking wave emitted from the internal core, providing diffusive acceleration.

Mass (m) is redefined as the **Adsorption Density**—the rigidity of the phase-alignment where these two waves synchronize in a $(1, 0, 1)$ routine.

1.2 Redefinition of Force (F) and Acceleration (a)

Force (F) is the **Phase Interference Pressure** required to perturb the anchored pattern. Acceleration (a) is the **Phase Transition Rate** (A_Ω) at which the pattern re-aligns to a new equilibrium.

$$F = m_{\text{pattern}} \times A_\Omega \tag{1}$$

2 Logarithmic Stacking in the 3-4-5 Lattice

2.1 The 3-4-5 Ratio as a Phase-Matching Threshold

The geometric stability of a 3-4-5 right triangle is shown to be a consequence of logarithmic phase-matching. In the Omega Axiom, energy transitions are non-linear and follow:

$$L_3 = \log(3), \quad L_4 = \log(4), \quad L_5 = \log(5) \tag{2}$$

The integers 3 (upper wave density) and 4 (lower wave density) converge into the anchored pattern of 5.

2.2 Force as the Gap between 3 and 4

The majority of observable force in the universe arises from the **Phase Gap** between the upper and lower waves.

- **Phase Gap**: The interval $|\log(4) - \log(3)|$ represents the potential energy of the pattern before it reaches the anchored state (5).
- By monitoring this gap, autonomous systems can determine collision trajectories with zero computational latency.

3 Engineering Application: 10-Layer Stacked Automata

The implementation of this theory in a 10-layer lattice allows for spatial orthogonality. Expanding from a $6 \times 6 \times 6$ core to a $25 \times 25 \times 25$ grid maintains deterministic stability, as the $(1, 0, 1)$ logic governs the transitions across the 3-4-5 logarithmic thresholds.